From discretuum to continuum

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I. INTRODUCTION

Regge calculus [1] offers a discrete formulation of classical general relativity (GR) on simplicial manifolds and plays a central role in discrete approaches to quantum gravity such as (causal) dynamical triangulations [2, 3], quantum Regge calculus [4, 5] and spin-foam models [6]. In spin-foams in particular, the Regge action appears in the large-spin regime of the vertex amplitude, and therefore serves as a semi-classical limiting point.

While originally formulated in terms of length variables and in Euclidean signature, recent developments highlight the importance of so-called area Regge calculus [7, 8] in Lorentzian signature. Area variables for Regge calculus are particularly interesting because of their direct connection to the quantum geometric degrees of freedom of spin-foams, loop quantum gravity [9, 10] and group field theories [11, 12]. Furthermore, recent renormalization group studies [13] in continuum area-metric gravity [14–16] suggests that GR is recovered from area variables in the infrared, i.e. at the coarse-grained level.

Area Regge calculus also forms the fundamental ingredient of effective spin-foam models [17–21] which are computationally feasible and facilitated studies concerning the causal structure [22], quantum cosmology [23], the de Sitter horizon entropy [24], and so-called spikes and spine configurations [25, 26]. Studies on the causal structure and the application of Regge calculus to cosmology are of particular interest here.

Explicitly Lorentzian configurations in the form of spacelike, lightlike and timelike building blocks have only recently gained attention in Regge calculus and related quantum gravity approaches. In [27, 28] importance of timelike building blocks on-shell, causality violations, effective spin-foams and suppression of causality violations

- Regge calculus important as it appears in the semi-classical limit of spin-foams
- close connection to effective spin-foams
- The continuum limit
- Lorentzian signature

$8\pi G_{\mathrm{N}} = 1$?, spacetime signature $(+,-,-,-)$

II. SETUP

A. Lorentzian Regge calculus

Regge calculus is a discrete gravitational theory on simplicial manifolds Δ with the length of edges $\{l_e\}$ dynamical. In a Lorentzian setting, a simplicial manifold consists of intrinsically flat Lorentzian 4-simplices $\sigma \in \Delta$. Curvature is located at triangles $t \in \Delta$ and captured by deficit angles $\delta_t(\{l_e\})$, lying in the space orthogonal to t. For t spacelike (timelike), this space is isomorphic to $\mathbb{R}^{1,1}$ ($\mathbb{R}^2$).[1] Lorentzian deficit angles are defined as $\delta_t^{\mathrm{L},\pm} = \mp i2\pi - \sum_{\sigma\supset t}\psi_{\sigma,t}^{\mathrm{L},\pm}$, following the conventions of [20, 29], where the $\psi_{\sigma,t}$ are dihedral angles. "$\pm$" indicates a choice of sign which becomes important for causally irregular configurations [19, 20, 29] discussed below. The bulk action of Lorentzian Regge calculus is given by [1, 20, 29]

$$\begin{aligned} S_{\mathrm{R}}[\{l_e\}] = &\sum_{t\in\Delta:\ \mathrm{tl}} |A_t(\{l_e\})|\delta_t^{\mathrm{E}}(\{l_e\}) \\ &+ \sum_{t\in\Delta:\ \mathrm{sl}} |A_t(\{l_e\})|\delta_t^{\mathrm{L},\pm}(\{l_e\})\,, \end{aligned} \tag{1}$$

with $|A_t|$ the absolute value of the triangle area. In the presence of a boundary, $\partial\Delta \neq \emptyset$, S_{R} attains boundary terms such that overall additivity is ensured. The equations of motion are given by $\frac{\partial S_{\mathrm{R}}}{\partial l_e} = 0$ supplemented by the Schläfli identity, $\sum_t |A_t| \frac{\partial \delta_t}{\partial l_e} = 0$.

Regge calculus can be formulated in terms of different variables [30, 31], with area variables [7, 8, 32] playing a particularly important role here due to their connection to LQG and spin-foams. The relation between area and length variables is in general not invertible, and area variables need to be further constrained to yield the dynamical equations of length Regge calculus [8, 20, 31]. However, in the symmetry reduced setting introduced below, the relation of area and length is globally invertible and the aforementioned constraints trivialize. For the remainder, we study the classical dynamics of area Regge calculus which will be used in future work as a comparison to results of the effective spin-foam path integral.

B. Lorentzian 4-frusta

Lorentzian 4-dimensional frusta are ideal to capture spatially flat, homogeneous and isotropic discrete geometries [28, 33]. The boundary of a 4-frustum, depicted in Fig. 1, contains two spacelike 3-cubes and six 3-frusta that are either spacelike or timelike. Gluing 4-frusta along boundary 3-frusta in spatial direction and along boundary 3-cubes in temporal direction, one obtains an

[1] For the remainder, we neglect lightlike edges, triangles and tetrahedra. See [29] for the definition of deficit angles associated to $(d-1)$-dimensional null cells.

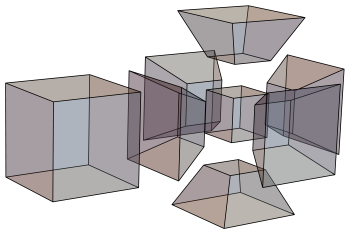

FIG. 1: Boundary of a 4-frustum given by two 3-cubes and six equal 3-frusta.

extended cosmological spacetime of hypercubical combinatorics with slices labelled by $n \in \{0, ..., \mathcal{V}_T\}$ discretized by $\mathcal{V}_S^3$ spacelike 3-cubes. We say that a 4-frustum lies in a "slab" between slices n and $n+1$.

The entire geometry of a slab is captured by three variables describing the 3-cubes and 3-frusta which can be either areas or lengths. The former set of variables consists of the area of squares contained in the 3-cubes, j_0 and j_1, and the area of the trapezoids contained in the 3-frusta, k. The latter consists of the edge length contained in the 3-cubes, s_0 and s_1, and the height of the 4-frustum, H_0, which implicitly determines the trapezoids and 3-frusta.[2] In the highly symmetry restricted setting considered here, there is a one-to-one map between area variables, (j_0, j_1, k_0), and length variables, (s_0, s_1, H_0), given explicitly in Appendix A. We will exploit this relation in Sec. III B using length variables for analytical computations and clear interpretation, especially because the height is directly related to the lapse function N in the continuum. When solving the Regge equations numerically thereafter, we will use area variables instead to provide a classical benchmark for the effective spin-foam calculations of [34]. The general notions of Regge calculus introduced above are straightforwardly adapted to either of the two sets of variables.

The squared volume of subcells connecting neighboring slices, i.e. 3-frusta, trapezoids and edges, which are detailed in Appendix A, carries information on the causal structure. If it is positive (negative), the corresponding subcell is timelike (spacelike). In the symmetry reduced setting, the ratio $\frac{|j_n - j_{n+1}|}{4k_n}$ fully characterizes the causal character of subcells suggesting separating the 4-frusta configurations into three sectors summarized in Tab. I.

As discussed in [21, 23, 27, 28, 35], two of these sectors I and II exhibit an irregular light cone structure, also referred to as causality violations, located at spacelike sub-cells. Examples of such irregularities are given by the trouser singularity and the yarmulke singularity associated to spatial topology change [36–38]. For causality violations at $(d-2)$-dimensional spacelike hinges the imaginary parts of the dihedral angles do not sum up to $\mp i2\pi$, leading to a complex valued Regge action. In effective spin-foams, choosing "+" in the convention of Lorentzian angles, the amplitude $\mathrm{e}^{iS_\mathrm{R}}$ is exponentially suppressed for trouser and exponentially enhanced for yarmulke configurations, vice versa for "−" [20].Without a complexification of the variables, the Regge equations of motion for causality violating configurations are complex equations with real variables and therefore overconstrained. Moreover, the results of [27, 28] show that in the respective settings, no real solutions to the equations of motion exist. Since this applies also to the setting used in this work we henceforth focus on the causally regular sector III.

C. Lorentzian Regge action for spatially flat cosmology

For a single 4-frustum between, say slice n and $n+1$, the causally regular Regge action is given by

$$S_{\mathrm{III}}^{(n)} = 6(j_n - j_{n+1}) \sinh^{-1}\left(\frac{j_{n+1} - j_n}{\sqrt{16k_n^2 + (j_n - j_{n+1})^2}}\right) + 12k_n\left[\frac{\pi}{2} - \cos^{-1}\left(\frac{(j_n - j_{n+1})^2}{16k_n^2 + (j_n - j_{n+1})^2}\right)\right] - \Lambda V^{(4)}\,, \qquad (2)$$

where Λ is a cosmological constant and $V^{(4)}$ is the 4-volume defined in Eq. (A4) as an equation of (j_n, j_{n+1}, k_n). By construction, the 4-frustum action is additive such that on a discretization with $\mathcal{V}_S^3$ vertices in spatial direction and $\mathcal{V}_T + 1$ spatial slices, the action is given by $S_\mathrm{R} = \mathcal{V}_S^3 \sum_{n=0}^{\mathcal{V}_T - 1} S_\mathrm{R}^{(n)}$. Additivity of the action per slab implies that the equations of motion for a given geometric quantity, being either j_n or k_n, will only depend on neighboring geometric labels.

In the following, we study the discrete dynamics of spatially flat cosmology as governed by the Regge equations. To that end, we consider first the case of a single time step, $\mathcal{V}_T = 1$, in Sec. III and generalize then to an arbitrary number of time steps in Sec. III B.

[2] Strictly speaking, the height H_0 is a derived quantity that is not a length associated to a 1-dimensional sub-cell of the cellular complex. Such a quantity would instead be given by the length l_0 associated to the strut connecting two neighboring spacelike slices. However, the relation between strut length and height is invertible for non-null struts, which is assumed here.

	Sector I	Sector II	Sector III.1	Sector III.2
sub-cells	$-\frac{\Delta j^2}{8} < \underline{k}^2 < -\frac{\Delta j^2}{16}$	$-\frac{\Delta j^2}{16} < \underline{k}^2 < 0$	$0 < \underline{k}^2 < \frac{\Delta j^2}{16}$	$\frac{\Delta j^2}{16} < \underline{k}^2$
struts	spacelike	spacelike	spacelike	timelike
trapezoids	spacelike	spacelike	timelike	timelike
3-frusta	spacelike	timelike	timelike	timelike

TABLE I: Causal characters of sub-cells as a function of (j_0, j_1, k), computed from the squared volumes given in Appendix A. Sectors III.1 and III.2 represent the hinge causally regular sector of the theory where solutions to the equations of motion exist.

III. RECAP

A. One time step

The classical and quantum dynamics of a single timestep in discrete Lorentzian cosmology have been studied extensively in [19, 23, 27] for spatial curvature with cosmological constant and in [21, 28] for spatially flat spacetimes with a free scalar field. In this section we extend the classical studies to spatially flat cosmology with a cosmological constant.

For $\mathcal{V}_T = 1$, the square areas j_0, j_1 are boundary data and only the trapezoid area k_0 is dynamical. Thus, the Regge equations are simply given by

$$\frac{\partial S_{\text{III}}}{\partial k_0} = 0\,, \tag{3}$$

where due to the Schläfli identity, the first term in Eq. (2) drops out. Notice that this equation is transcendental and can therefore only by tackled numerically.

Vanishing cosmological constant. In the case of $\Lambda = 0$, the Regge action describes a spatially flat vacuum universe. If the boundary data is non-stationary, i.e. $j_0 \neq j_1$, Eq. (3) exhibits no solution for k_0. In other words, a classical universe can neither expand nor contract if there is no matter present. This is in accordance with the continuum Friedmann equations in vacuum, $\left(\frac{\dot{a}}{a}\right)^2 = 0$, with a the scale factor, which are solved only by $a = \text{const}$. On the other hand, if $j_0 = j_1$, Eq. (3) is trivially satisfied and the trapezoid area k_0 is undetermined. In fact, the Regge action vanishes for all k_0, and the 4-frusta reduce to 4-cuboids discretizing flat space. The arbitrariness of k is a common phenomenon for flat solutions in Regge calculus [39, 40] and signifies the restoration of diffeomorphism symmetry in the discrete [41, 42]. This symmetry amounts to moving spacelike slices (instead of individual vertices), see also the discussion in [43]. Note that for 4-cuboids, $k = \sqrt{j}H$ with H the height of the cuboid which is directly related to the lapse function N in the continuum [27, 28]. The lapse is the gauge variable of the residual diffeomorphism symmetry after symmetry reduction, encoding time reparametrization invariance.

Non-vanishing cosmological constant. For a non-vanishing cosmological constant, $\Lambda \neq 0$, and stationary boundary conditions, $j_0 = j_1$, the only solution to Eq. (3) is $k_0 = 0$, corresponding to degenerate 4-frusta with vanishing height. Thus, a stationary universe for $\Lambda \neq 0$ only exists if no time passes, i.e. if there is no actual time evolution. The more interesting case is $j_0 \neq j_1$ where a non-trivial solution k_0^* for Eq. (3) exists. More precisely, a unique solution k_0^* is found for any $\Lambda > 0$ and any $j_0 \neq j_1$. The on-shell trapezoid area $k_0^*(\Lambda)$ as a function of Λ behaves as a monomial decay $k_0^* \sim c\Lambda^{-\alpha}$ with $c, \alpha > 0$ constants depending on j_0, j_1. For $\Lambda \neq 0$, the solution diverges and ceases to exist at $\Lambda = 0$ in lines with the discussion above. In the limit $\Lambda \to \infty$, Eq. (3) is dominated by the Λ-term, i.e.

$$\Lambda \frac{(j_0 + j_1)^2 k}{4V^{(4)}} \approx 0\,, \tag{4}$$

solved by $k_0^* = 0$.

Note that the restriction on the sign of Λ is in accordance with the continuum Friedmann equations, $\left(\frac{\dot{a}}{a}\right)^2 = N^2 \frac{\Lambda}{3}$, where real solutions for the scale factor require $\Lambda > 0$.

For completeness, let us contrast these results with the existence of solutions in the spatially spherical case in Lorentzian [27] and Euclidean [44, 45] signature.[3] In the former setting, note first that for $\Lambda = 0$, no solutions exist. Furthermore, for $\Lambda > 0$, solutions exist only if $j_0 > j_\Lambda$ and $j_1 > j_\Lambda$, where j_Λ is a Λ-dependent minimal area. This is in accordance with the continuum where the scale factor typically follows a cosh-function, attaining a minimal value a_Λ. In Euclidean signature, solutions exist for $\Lambda = 0$. These correspond, in the continuum, to solutions where $\dot{a} = \text{const}$. For $\Lambda > 0$, solutions exist only if $j_0 < j_\Lambda$ and $j_1 < j_\Lambda$ where j_Λ is the same as

[3] The construction of the spatially spherical Regge action differs from the construction here, in particular because there exist only finitely many homogeneous triangulations of the 3-sphere, see [27, 45] for details.

in the Lorentzian case, however now representing an upper bound. Again, this behavior agrees with continuum results where a Euclidean universe with positive spatial curvature can only reach a maximum value a_Λ.

B. Arbitrary number of time steps

In this section, we study the Regge equations of spatially flat Lorentzian cosmology for an arbitrary number of time steps $\mathcal{V}_T$. This amounts to solving $2\mathcal{V}_T - 1$ coupled transcendental equations for the bulk variables $k_0, j_1, \ldots, j_{\mathcal{V}_T-1}, k_{\mathcal{V}_T-1}$ for given boundary data $j_0, j_{\mathcal{V}_T}$ and cosmological constant Λ,

$$\frac{\partial S_{\text{III}}}{\partial k_n} = 0\,, \qquad \frac{\partial S_{\text{III}}}{\partial j_m} = 0\,, \tag{5}$$

with $n \in \{0, \ldots, \mathcal{V}_T - 1\}$ and $m \in \{1, \ldots, \mathcal{V}_T - 1\}$. Since the Regge action S_{III} is additive with respect to the slices, i.e. $S_{\text{III}} = \mathcal{V}_S^3 \sum_n S_{\text{III}}^{(n)}$, Eqs. (5) simplify significantly,

$$\begin{aligned} 0 &= \frac{\partial S_{\text{III}}}{\partial k_n} = \mathcal{V}_S^3 \frac{\partial S_{\text{III}}^{(n)}}{\partial k_n}\,, \\ 0 &= \frac{\partial S_{\text{III}}}{\partial j_n} = \mathcal{V}_S^3 \left(\frac{\partial S_{\text{III}}^{(n-1)}}{\partial j_n} + \frac{\partial S_{\text{III}}^{(n)}}{\partial j_n} \right). \end{aligned} \tag{6}$$

Thus, it is sufficient to consider the single slab action $S_{\text{III}}^{(n)}$, given in Eq. (3), and its derivatives. An explicit form of these equations is given in Appendix A.

A priori, the number of cubes in each spacelike slice, $\mathcal{V}_S^3$, enters the total Regge action only as a multiplicative factor due to the symmetry restriction. Thus, in the following two sections, we consider $\mathcal{V}_S = 1$ for simplicity. However, let us mention here that in Sec. V, the parameter $\mathcal{V}_S$ turns out to be crucial to obtain the correct continuum limit. We tackle the problem of finding solutions to Eq. (5) first analytically in the subsequent section and develop numerical methods thereafter.

For vanishing cosmological constant, $\Lambda = 0$, and stationary boundary data, $j_0 = j_{\mathcal{V}_T}$, the results for a single time step obtained in the previous section straightforwardly generalize. That is, if $\Lambda = 0$, the Regge equations are satisfied if and only if the boundary data satisfies $j_0 = j_{\mathcal{V}_T}$. Then, the bulk areas $j_n = j_0$ for all $n \in \{1, \ldots, \mathcal{V}_T - 1\}$, and the trapezoid areas are undetermined, marking the restoration of time reparametrization invariance of flat space. On the other hand, for $\Lambda \neq 0$ and boundary data $j_0 = j_{\mathcal{V}_T}$, no non-degenerate solution to the Regge equations is found. A unique solution exist for $j_0 \neq j_{\mathcal{V}_T}$ when $\Lambda > 0$ which is in accordance with the continuum Friedmann equations. To summarize, a spatially flat vacuum universe is flat and stationary while a positive cosmological will necessarily lead to curved non-stationary solutions. For the remainder, we assume $\Lambda > 0$ and $j_0 \neq j_{\mathcal{V}_T}$.

Eq. (5) has been analyzed further in [28] (coupling a scalar field rather than a cosmological constant). Therein, the length variables (s_n, s_{n+1}, H_n) have been employed in which the monotonicity of solutions and the continuum limit are derived most easily. The results of [28] transferred to the present setting with $\Lambda \neq 0$ are briefly summarized in the following.

Combining the Regge equations

$$\frac{\partial S_{\text{III}}}{\partial H_{n-1}} = 0\,, \quad \frac{\partial S_{\text{III}}}{\partial H_n} = 0\,, \quad \frac{\partial S_{\text{III}}}{\partial s_n} = 0\,, \tag{7}$$

in the variables (s_n, s_{n+1}, H_n), one can straightforwardly derive the relation [28]

$$\frac{s_n - s_{n-1}}{H_{n-1}} = \frac{s_{n+1} - s_n}{H_n}\,, \tag{8}$$

for all $n \in \{1, \ldots, \mathcal{V}_T - 1\}$. That is, the difference of spatial edge length with respect to the height of the 4-frusta remains constant. Introducing

$$\tau_{\mathcal{V}_T} = \sum_{n=0}^{\mathcal{V}_T-1} H_n\,, \tag{9}$$

which we will later identify as the discrete elapsed proper time, one can show that Eq. (8) implies

$$\frac{s_{n+1} - s_n}{H_n} = \frac{s_{\mathcal{V}_T} - s_0}{\tau_{\mathcal{V}_T}}\,, \tag{10}$$

for all n. Since $H_n, \tau_{\mathcal{V}_T} > 0$, this implies $\mathrm{sgn}(s_{n+1} - s_n) = \mathrm{sgn}(s_{\mathcal{V}_T} - s_0)$, that is, solutions evolve monotonically and the boundary data $s_0, s_{\mathcal{V}_T}$ selects whether the universe is expanding, $s_{\mathcal{V}_T} > s_0$, or contracting, $s_{\mathcal{V}_T} < s_0$.

Following the procedure detailed in [28], an analytical continuum limit of the Regge equations can be taken in two steps. First, one performs a linearization of $\frac{\partial S_{\text{III}}}{\partial H_n} = 0$ in the variable $\eta_n = \frac{s_{n+1} - s_n}{H_n}$. In this approximation, corresponding to small deficit angles, one finds

$$\frac{(s_n - s_{n+1})^2}{s_n^2 + s_{n+1}^2} = H_n^2 \frac{\Lambda}{6}\,. \tag{11}$$

Then, the discrete variables (s_n, s_{n+1}, H_n) need to be identified with the continuum scale factor $a(t)$ and the lapse function $N(t)$ parametrized in some coordinate time t. In the continuum limit, the height H_n corresponds to the infinitesimal proper time, $H_n \to N\mathrm{d}t$. The discrete length on a given slice is mapped to the scale factor, $s_n \to a(t)$, while the length of a neighboring edge attains an additional time derivative term, $s_{n+1} \to a(t) + \dot{a}\mathrm{d}t$. Then, at leading order in $\mathrm{d}t$, on finds indeed that Eq. (11) approaches

$$\left(\frac{\dot{a}}{a}\right)^2 = N^2 \frac{\Lambda}{3}\,, \tag{12}$$

which is the Friedmann equation for a spatially flat universe with cosmological constant. Note that configurations with spacelike trapezoids or spacelike 3-frusta satisfy $\eta_n^2 > 2$ and therefore exhibit no straightforward continuum limit [28].

The cosmological Regge equations have been shown to capture the Friedmann equations in a suitable continuum limit in previous works [27, 28, 33, 46]. Still, Eq. (5) remains to be solved explicitly in the discrete for many time steps $\mathcal{V}_T$. This represents an important step since the solutions serve as a classical and discrete benchmark for identifying a semi-classical continuum regime in discrete path integral approaches such as effective spin-foams. In the following section, we tackle this issue employing numerical methods.

IV. NUMERICAL IMPLEMENTATION

Consider turning the numerics section into its own section and putting the stuff before in the recap section. After all, the stuff before has already been discussed in [28].

Solving the Regge equations explicitly requires numerical methods. In this work, we employ the package `NLsolve`[4] in `Julia`, designed for solving non-linear equations. Note that while Eq. (5) defines real-valued equations in real variables, `NLsolve` can also be applied to complex numbers which may be relevant for the complex critical point method []. The algorithm we developed for this work specifically tailored to Eq. (5) amounts to the `solve_eom`[5]-method, based on the `nlsolve`-function of `NLsolve`, which we explain in the following.

First, note that `nlsolve` takes as input a vector valued function $\vec{F}$, a vector of initial variables $\vec{x}_0$, and parameters for the number of iterations and the error tolerance for solutions. If the problem is sufficiently well-behaved, `nlsolve` finds a solution to the equation $\vec{F}(\vec{x}) = \vec{0}$ using $\vec{x}_0$ as a starting guess. In our case, $\vec{F} : \mathbb{R}^{2\mathcal{V}_T - 1} \longrightarrow \mathbb{R}^{2\mathcal{V}_T-1}$, with

$$\vec{F} = \left(\frac{\partial S_{\text{III}}}{\partial k_0}, \frac{\partial S_{\text{III}}}{\partial j_1}, \dots, \frac{\partial S_{\text{III}}}{\partial j_{\mathcal{V}_T - 1}}, \frac{\partial S_{\text{III}}}{\partial k_{\mathcal{V}_T-1}} \right) , \tag{13}$$

$$\vec{x} = (k_0, j_1, \dots, j_{\mathcal{V}_T - 1}, k_{\mathcal{V}_T -1}) \; . \tag{14}$$

Therefore, Eq. (5) is adapted to `nlsolve` in two steps: 1) set up the function $\vec{F}$ for fixed boundary areas $j_0, j_{\mathcal{V}_T}$ and cosmological constant Λ, and 2) propose a starting guess $\vec{x}_0$. While step 1) is straightforward, step 2) requires particular care because the starting guess $\vec{x}_0$ is crucial for convergence and good performance of the solving algorithm. In particular, as $\mathcal{V}_T$ grows, the number of variables and equation grows and the algorithm becomes more sensitive to the choice of $\vec{x}_0$.

By trial and error, one can find sufficient starting guesses for `nlsolve` to converge given various small $\mathcal{V}_T$.[6] For the solutions $\vec{x}^* = (k_0^*, j_1^*, \dots, j^*_{\mathcal{V}_T - 1}, k^*_{\mathcal{V}_T - 1})$, the following patterns are being observed. First, the j_n^* can be approximated as

$$j_n^* \approx j_0 + n\Delta j \, , \quad n \in \{1, \dots, \mathcal{V}_T - 1\} \, , \tag{15}$$

with $\Delta j = (j_{\mathcal{V}_T} - j_0)/\mathcal{V}_T$. Second, let $\boldsymbol{k}_0^*$ be the solution of the trapezoid area for $\mathcal{V}_T = 1$. Then, for $\mathcal{V}_T > 1$, the k_n^* are approximated by

$$k_n^* \approx \boldsymbol{k}_0^* / \mathcal{V}_T \, , \quad n \in \{0, \dots, \mathcal{V}_T - 1\} \, . \tag{16}$$

That is, to find a good initial guess for the trapezoid areas, one only needs to find a solution $\boldsymbol{k}_0^*$ for a single time step. Crucially, for a single equation, `nsolve` converges for a broad range of initial guesses, so $\boldsymbol{k}_0^*$ can be found easily. In summary, finding a good initial guess $\vec{x}_0$ amounts to finding the one time step solution $\boldsymbol{k}_0^*$ and then setting

$$\vec{x}_0 = \left(\frac{\boldsymbol{k}_0^*}{\mathcal{V}_T}, j_0 + \Delta j, \dots, j_0 + (\mathcal{V}_T - 1)\Delta j, \frac{\boldsymbol{k}_0^*}{\mathcal{V}_T} \right) \, . \tag{17}$$

Implementing this setup, the function `solve_eom` takes as input the boundary data $j_0, j_{\mathcal{V}_T}$, the cosmological constant Λ, a parameter $\gamma = 1$[7], and solver-specific arguments such as the maximum number of iterations or error tolerances. The output of `solve_eom` is a tuple consisting of four arrays: the $\mathcal{V}_T + 1$ square area solutions j_n^*, including the boundary values, the $\mathcal{V}_T$ trapezoid areas k_n^*, the heights of 4-frusta H_n^* and the accumulated heights $\tau_n = \sum_{k=0}^{n-1} H_k^*$.

In the following, we discuss the numerical stability and time complexity of the algorithm before then applying `solve_eom` to study the properties of discrete solutions and their continuum limit.

A. Numerical stability and time complexity

Performance of the solving algorithm is not only measured by its ability to find solutions but also by its stability and its time efficiency. As discussed in the previous section, the initial guess $\vec{x}_0$ is crucial for the performance of `nlsolve`. Therefore, we will check first, whether `solve_eom` is stable under perturbations $\vec{\delta}$ of $\vec{x}_0$. Second, we estimate the scaling of computation time with the number of time steps $\mathcal{V}_T$. This is particularly important for later studies of the continuum limit, where access to large $\mathcal{V}_T$ is of physical interest.

[4] A documentation of the package can be found at github.com/JuliaNLSolvers/NLsolve.jl. Note that also the `NSolve` method in `Mathematica` can be used, but is expected to have lower performance in terms of accuracy and computation time.

[5] See this github repository for the source code.

[6] We tried using the continuum solution to find a good initial guess. However, we found that the prescription given here ensures fast convergence more reliably.

[7] In future work, when comparing effective spin-foam results with the classical solutions, $\gamma \in \mathbb{R}$ will be allowed to vary, corresponding to the Barbero-Immirzi paramter [].

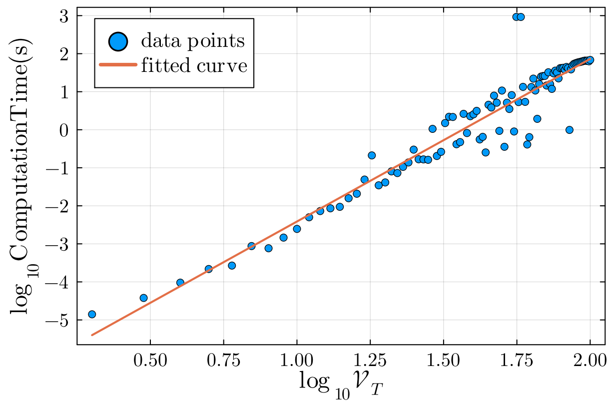

FIG. 2: Logarithm of computation time as a function of $\log \mathcal{V}_T$. The fit via linear regression yields $t(V) = 10^{\hat{\alpha}} V^{\hat{\beta}}$ with $\hat{\alpha} = -6.67$ and $\hat{\beta} = 4.23$. For demonstrative purposes, the boundary data and Λ have been fixed to $j_0 = 50, j_{\mathcal{V}_T} = 100$ and $\Lambda = 0.1$.

To understand the behavior of the solver under perturbations of the initial guess $\vec{x}_0$, consider a vector $\vec{\delta} = \frac{1}{\sqrt{2\mathcal{V}_T - 1}}(\delta, \dots, \delta) \in \mathbb{R}^{2\mathcal{V}_T - 1}$. Then, `solve_eom` with initial guess $\vec{x}_0 + \vec{\delta}$ is evaluated, the results of which are discussed separately for the cases if $\mathcal{V}_T = 1$ and $\mathcal{V}_T > 1$.

$\mathcal{V}_T = 1$. `solve_eom` yields consistent results as long as the entries of $\vec{x}_0 + \vec{\delta}$ are positive. For negative entries, there are non-trivial sign changes in the Regge equation, and in this case, it not clear if solutions even exist. Choosing δ such that the entries are positive, the solutions are stable even for large δ that are the order of the boundary values $j_0, j_{\mathcal{V}_T}$. Also, the number of iterations does only increase slightly. This is important since setting up $\vec{x}_0$ requires finding $\boldsymbol{k}_0^*$ first. If this first step would be extremely sensitive to the initial guess, the proposal above in Eq. (17) would not be viable.

$\mathcal{V}_t > 1$. In this case, the number of iterations needed to find convergence is generically larger, even for $\delta = 0$. Choosing δ such that the entries of $\vec{x}_0 + \vec{\delta}$ are positive, one finds that solutions are stable for small $|\delta|$ with little fluctuations in the required iterations. However, for larger δ, the solutions suddenly jump to unreasonably large or to negative values and the number of iterations increases. This effect worsens with $\mathcal{V}_T$. For the solver, this means that 1) the proposal of $\vec{x}_0$ given above is already optimal and one should stick with it, and 2) for very large $\mathcal{V}_T$, the region of trustworthy convergence might shrink to zero and one needs to come up with a more refined proposal of initial guesses than $\vec{x}_0$ in Eq. (17). In this work, the values we consider for $\mathcal{V}_T$ are large but still small enough for $\vec{x}_0$ to yield satisfactory results.

To measure how computation time scales with the number of time steps, $\mathcal{V}_T$ (and thus with the number of equations, $2\mathcal{V}_T - 1$), we fix the boundary data $j_0, j_{\mathcal{V}_T}$ and Λ and apply `solve_eom` with increasing $\mathcal{V}_T$. In the log-log-plot of Fig. 2, one observes a monomial scaling. Performing a linear regression, one find that the computation time as a function $\mathcal{V}_T$ can be estimated as $t(\mathcal{V}_T) = 10^{\hat{\alpha}} V^{\hat{\beta}}$ with $\hat{\alpha} = -6.67$ and $\hat{\beta} = 4.23$. That is, the computation time for small $\mathcal{V}_T$ is very small, $\sim 10^{-6}$s, but increases as a fourth-order polynomial in $\mathcal{V}_T$. Still, the times at $\mathcal{V} \sim 10^2$ are reasonably small so that the behavior of solutions in the continuum limit, tackled in the next section, can be studied without any computation time issues.

V. CONTINUUM-INFORMED DISCRETIZATION

A. Identifying the continuum

Facilitated by the numerical tools developed above, we study in this section the continuum limit of Regge calculus at the level of the solutions, rather than at the level of the equations. That is, we solve the Regge equations explicitly and study their behavior when increasing the number of building blocks. This approach to the continuum limit is new and has not been considered in previous studies, where instead, the analytical continuum time limit similar to Sec. **??** has been performed.

Quantifying whether the discrete solutions provide a satisfactory approximation to the continuum ones requires two ingredients: 1) appropriate continuum quantities which serve as limiting points of the discrete solutions, and 2) identifying the right parameters to tune. Here, the discreteness is parametrized by the number of slices, $\mathcal{V}_T$, as well as the number of 3-cubes per slice, $\mathcal{V}_S^3$. Thus, the continuum is expected to be reached by scaling $\mathcal{V}_T, \mathcal{V}_S \to \infty$ in a suitably combined way.

1) Continuum quantities. One difficulty in comparing discrete and continuum solutions is that in the discrete, the trapezoid areas are dynamical while in the continuum, the lapse function N is a gauge variable, and thus, any choice of N corresponds to a time parametrization. Therefore, the following question arises: when performing the continuum limit, which time parametrization in the continuum emerges? Clearly, the labels $n \in \{0, \dots, \mathcal{V}_T\}$ do not serve as a time as this would imply constant temporal separation between any two neighboring spacelike slices. Instead, recall that the height of 4-frusta was mapped to infinitesimal proper time in the analytical continuum limit of Sec. **??**. Therefore, the sum of heights, $\tau_{\mathcal{V}_T} = \sum_n H_n$, serves as a discrete approximation of the continuum proper time elapsed, defined as $\int \mathrm{d}t\, N(t)$. The advantage of considering proper time is that it is gauge invariant. Thus, one does not run into the issue of having to pick a parametrization in the continuum to compare to the discrete solutions. Using proper time, we will then compare the evolution of the square area as a function of the accumulated height $\tau_n = \sum_{k=0}^{n-1} H_k$, i.e. the pair (τ_n^*, j_n^*), with the squared scale factor $A = a^2$ as a function of proper time, $(\tau_{\mathrm{cont}}, A(\tau))$.

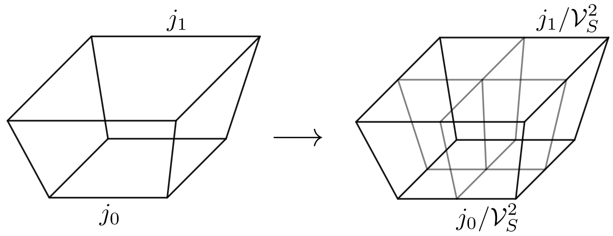

FIG. 3: Refinement of frustum geometry with total areas j_0 and j_1. Individual frusta are rescaled as $(j_0, j_1) \to (j_0/\mathcal{V}_S^2, j_1/\mathcal{V}_S^2)$. For illustrative purposes, one spatial dimension is suppressed.

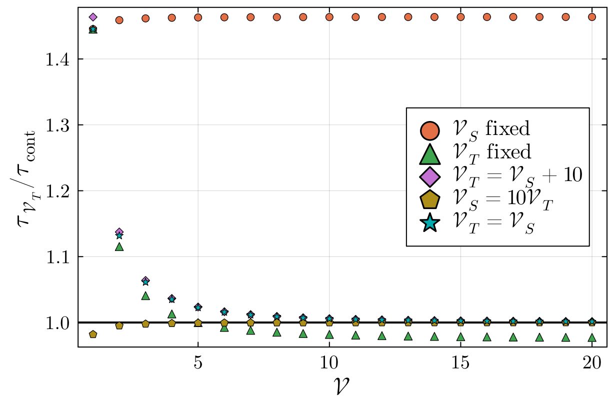

FIG. 4: Ratio of discrete and continuum proper time, τ_0 and τ_{cont}, respectively, as a function of the number of building blocks, $\mathcal{V}$. $\mathcal{V}_T$ and $\mathcal{V}_S$ are related as indicated. The boundary data and Λ have been fixed to $j_0 = 50$, $j_{\mathcal{V}_T} = 100$, and $\Lambda = 0.1$.

2) The right scaling. A priori, the parameter $\mathcal{V}_S$ enters the Regge action only as a multiplicative factor. Therefore, it might be tempting to set $\mathcal{V}_S = 1$, as done in the analytical part in Sec. **??**. Note, however, that when describing the same total boundary area $j_0, j_{\mathcal{V}_T}$ with a different number of spacelike building blocks, a re-scaling in the 4-frustum Regge action is required,

$$S_{\text{III}}^{(n)}(j_n, j_{n+1}, k_n) \to \mathcal{V}_S^3 S_{\text{III}}^{(n)}(j_n/\mathcal{V}_S^2, j_{n+1}/\mathcal{V}_S^2, k_n)\,. \quad (18)$$

This spatial refinement is depicted in Fig. 3. Importantly, the Regge action and thus also the solutions, are in general not invariant under the procedure above. Therefore, different combined scalings of $\mathcal{V}_S, \mathcal{V}_T \to \infty$ will a priori lead to different limiting points.

To determine which of those points corresponds to the continuum, the ratio of discrete and continuum proper time, $\tau_{\mathcal{V}_T}/\tau_{\text{cont}}$, is plotted in Fig. 4 as a function of building blocks $\mathcal{V}$. Clearly, continuum proper time is reached when this ratio equals 1, indicated by the black horizontal line. One observes fast convergence, already at $\mathcal{V} \gtrsim 10$ for the given boundary data and Λ to a limiting value. Note that the speed of convergence depends on the boundary data and will decrease for larger curvature, that is larger difference of j_0 and $j_{\mathcal{V}_T}$. Evidently, keeping either $\mathcal{V}_S$ or $\mathcal{V}_T$ fixed yields a non-vanishing deviation to the continuum proper time. Also other scalings where the ratio $\mathcal{V}_S/\mathcal{V}_T$ does not approach a constant value (such as $(\mathcal{V}_T, \mathcal{V}_S) = (\mathcal{V}, \mathcal{V}^2)$ with $\mathcal{V} \to \infty$) lead to such deviations. Therefore, in order to approach continuum proper time, the pair $(\mathcal{V}_T, \mathcal{V}_S)$ must be scaled as $(\mathcal{V}, \kappa\mathcal{V} + \text{const.})$ with $\mathcal{V} \to \infty$, and κ a constant. The constant shift can be set to zero as it becomes irrelevant for $\mathcal{V} \gg 1$ and slows down convergence, see Fig. 4.

Choosing $\kappa = 1$ for the moment, discrete and continuum solutions are depicted in Fig. 5 for some chosen boundary data and Λ. We observe that for increasing $\mathcal{V}$, the discrete solutions get closer to the continuum. Furthermore, mote that the deviations in proper time add up. This happens because the discrete proper time is a sum of heights, all of which show deviations to continuum proper time steps for small $\mathcal{V}$. For sufficiently large $\mathcal{V}$, e.g. at $\mathcal{V} = 10$ for the data of Fig. 5, the discrete solutions become a discretization of the continuum solutions.

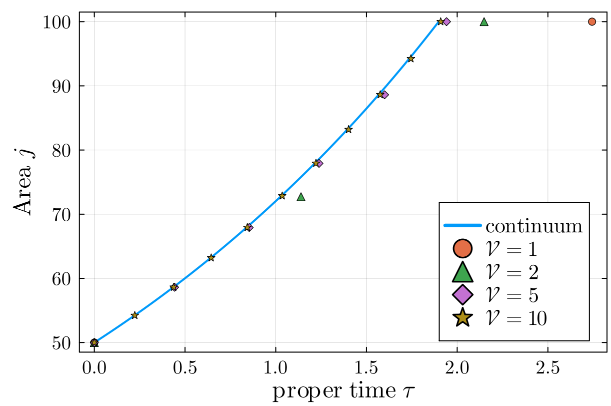

FIG. 5: Continuum and discrete area of spacelike slices as a function of proper time τ. The boundary data and Λ have been fixed to $j_0 = 50$, $j_{\mathcal{V}_T} = 100$, and $\Lambda = 0.1$.

In summary, we have identified proper time and the squared scale factor as a function of proper time as limiting points of the discrete solutions. Furthermore, we saw that the number of timelike and spacelike building blocks needs to be scaled as $(\mathcal{V}_T, \mathcal{V}_S) = (\mathcal{V}, \kappa\mathcal{V})$ with $\mathcal{V} \to \infty$ to reach the continuum. The convergence of (τ_n^*, j_n^*) to $(\tau, a^2(\tau))$ was demonstrated in Fig. 5.

These results show for the first time how, at the level of solutions, discrete Regge calculus explicitly converges to continuum gravity, facilitated by the numerical solver developed in Sec. III B. In the next section, we go one step further and show that there is in fact a choice of κ, relating $\mathcal{V}_T$ and $\mathcal{V}_S$, for which already at $\mathcal{V}_T = 1$, the continuum is perfectly captured.

B. Continuum-informed discretization

The previous section revealed that $(\mathcal{V}_T, \mathcal{V}_S)$ need to be scaled as $(\mathcal{V}, \kappa\mathcal{V})$ with $\mathcal{V} \to \infty$ to reach the continuum. Fig. 4 demonstrates that κ crucially influences the speed of convergence. Therein, the brown pentagons, representing $\kappa = 10$, clearly converge faster to $\tau_0/\tau_{\text{cont}} = 1$ than the blue stars with $\kappa = 1$. If we allow for $\mathcal{V}_S \in \mathbb{R}$ rather than being integer-valued[8], and therefore consider $\kappa \in \mathbb{R}$, we can ask whether there is a κ_* for which the continuum is obtained already at $\mathcal{V} = 1$.

This question can be formulated as an equation in κ for which then, if existent, κ_* is the solution. To that end, notice that the continuum is reached when $\tau_{\mathcal{V}_T} = \sum H = \tau_{\text{cont}}$. But if $\mathcal{V}_T = 1$, this is equivalent to

$$H(j_0/\kappa^2, j_1/\kappa^2, k) = \tau_{\text{cont}}\,, \tag{19}$$

that is, the elapsed proper time is equivalent to the height of the single frustum. By inverting the relation between k and H, see Eq. (A1), one can thus consider the continuum-informed trapezoid area

$$k_\tau(j_0, j_1, \kappa) = k(j_0/\kappa^2, j_1/\kappa^2, \tau_{\text{cont}})\,. \tag{20}$$

Then, the Regge equation $\frac{\partial S_{\text{III}}}{\partial k} = 0$ can be reformulated as an equation in κ, that is

$$12\left[\frac{\pi}{2} - \cos^{-1}\left(\frac{(j_0 - j_1)^2}{16k_\tau^2\kappa^4 + (j_0 - j_1)^2}\right)\right] - \sqrt{2}\Lambda\frac{(j_0 + j_1)k_\tau}{\sqrt{8k_\tau^2\kappa^4 + (j_0 - j_1)^2}} = 0\,. \tag{21}$$

Note that k_τ also implicitly depends on κ and that this equation is still transcendental.

The numerical methods developed in Sec. IV can be straightforwardly adjusted to find solutions of Eq. (21). Indeed, one can find a unique solution κ_* for any $j_0 \neq j_1$ and $\Lambda > 0$. A good initial guess for the solver can be looked for graphically by creating a plot similar to Fig. 4 and varying κ until all points lie on the horizontal line at $\tau_0/\tau_{\text{cont}} = 1$. The dependence of the solution κ_* on the boundary data and Λ is: 1) For fixed boundary data, κ_* depends on the square root of the cosmological constant, that is, $\kappa_*(\Lambda) = \beta\sqrt{\Lambda}$ for some β. 2) For fixed Λ, re-scaling the boundary as $(j_0, j_1) \to (\lambda j_0, \lambda j_1)$, the solutions of Eq. (21) scale as $\kappa_*(\lambda) = \beta'\sqrt{\lambda}$. Points 1) and 2) imply that for sufficiently small Λ or λ, κ_* becomes smaller than one and so $\mathcal{V}_T > \mathcal{V}_S$. 3) Fixing Λ and varying the difference $\Delta j = j_1 - j_0$, κ_* follows a monomial decay to some non-zero value as $\Delta j \gg 1$, that is, $\kappa_*(\Delta j) = (\Delta j)^{-\alpha} + c$ for some $\alpha > 0$ and some $c > 0$.

Equation (19) describes the identification of the frustum height with the continuum proper time. Inserting this continuum quantity and solving Eq. (21) for the ratio of building blocks in spatial and temporal direction therefore yields a *continuum-informed* discretization. Once κ_* is found, it can be inserted into the solver of Sec. IV so that then, the discrete solutions are a sampling of the continuum, as depicted in Fig. 6. Increasing $\mathcal{V}$ then simply corresponds to a finer sampling. Remarkably, that can be achieved by solving the single, though transcendental, equation (21) in contrast to the usual difficulty of obtaining continuum gravitational physics from discrete theories.

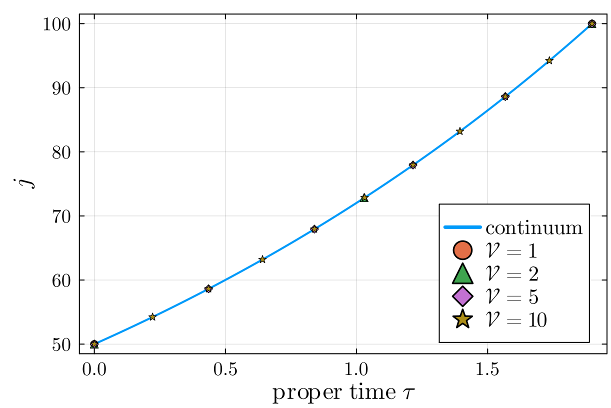

FIG. 6: Continuum and discrete area of spacelike slices as a function of proper time τ with $(\mathcal{V}_T, \mathcal{V}_S) = (\mathcal{V}, \kappa_*\mathcal{V})$. Evidently, increasing $\mathcal{V}$ only leads to a finer sampling of the continuum solution. The boundary data and Λ have been fixed to $j_0 = 50$, $j_{\mathcal{V}_T} = 100$, and $\Lambda = 0.1$.

Generically, the solutions κ_* will be real-valued, although there exist fine-tuned j_0, j_1 and Λ such that κ is an integer. Allowing $\kappa \in \mathbb{R}$ is only possible in the present setting of spatially flat and homogeneous cosmology. As soon as spatial flatness or homogeneity are dropped, $\mathcal{V}_S$, and thus κ, can only take integer values. In the spatially spherical case [27] in particular, the number of tetrahedra is fixed to a small subset of integers and cannot be simply played around with. Even if one considers non-integer κ as unphysical, the discussion above is helpful to find continuum solutions for small $\mathcal{V}$. To that end, one simply considers $\text{int}(\kappa)$, i.e. the integer closest to κ, so that convergence to the continuum in the limit $(\mathcal{V}_T, \mathcal{V}_S) = (\mathcal{V}, \kappa\mathcal{V})$ with $\mathcal{V} \to \infty$ will be as fast as possi-

[8] Note that this is only possible in the present symmetry restricted setting. As soon as the assumption of spatial homogeneity is dropped, $\mathcal{V}_S$ determines the number of spatial variables and can therefore not be treated anymore like some tunable real parameter. We elaborate on this point again down below.

ble.

C. A perfect action?

The existence of κ_* for the present curved discrete geometries is remarkable as it represents a form of *fixed point* of the limit of increasing building blocks, $\mathcal{V} \to \infty$. That is, evaluated on κ_*, the Regge action and its solutions become independent of $\mathcal{V}$ and the discretization merely corresponds to a sampling of the continuum. Thus, already at a single time step $\mathcal{V}_T = 1$, the Regge action is a perfect action, capturing the continuum dynamics without any discreteness artifacts. There are only few simple examples of discretized theories for which perfect actions have been found, including the free particle [], the harmonic oscillator [] and Regge calculus for flat spacetime []. Our results here represent the first example of Regge calculus with curvature where the perfect action can be found explicitly.

- The action is not a perfect action
- effective action just scales with $\lambda^{3/2}$.
- only one equation required for κ.

Continuum limit also in the analytical sense? Consider $s_n \to s_n/\mathcal{V}_S$.

VI. CONCLUSION

Extension to $k = \pm 1$.

Acknowledgements

Appendix A: Frusta geometry

In area variables, the geometry of a slab is fully characterized by (j_0, j_1, k), the square area of the cubes in slice 0 and 1, respectively, and the trapezoid area k. We provide the formulae of the squared volumes of relevant sub-cells in the following. Here, the square areas j_0, j_1 are assumed to be the absolute values of the signed areas and therefore positive. For the squared trapezoid area, we write $\underline{k}^2$ for the signed squared area and k^2 for its absolute value.

A crucial quantity is the squared height of the 4-frusta,

$$H^2 = \frac{8\underline{k}^2 + (j_0 - j_1)^2}{2(\sqrt{j_0} + \sqrt{j_1})^2}\,, \tag{A1}$$

due to its direct relation to the proper time in the continuum. Furthermore, it gives a bound $8\underline{k}^2 > -(j_0 - j_1)^2$ for the 4-frusta to be embeddable in Minkowski space. Configurations that violate this bound correspond to Euclidean frusta.

The squared length of struts, which connect neighboring spacelike slices is

$$l^2 = \frac{16\underline{k}^2 - (j_0 - j_1)^2}{4(\sqrt{j_0} + \sqrt{j_1})^2}\,. \tag{A2}$$

Thus, configurations with $16\underline{k}^2 > (j_0 - j_1)^2$ have timelike struts. If equality holds instead, struts are lightlike. Otherwise, they are spacelike. The squared height of 3-frusta is

$$h^2 = \frac{16\underline{k}^2 + (j_0 - j_1)^2}{4(\sqrt{j_0} + \sqrt{j_1})^2}\,, \tag{A3}$$

so 3-frusta are timelike if $16\underline{k}^2 > -(j_0 - j_1)^2$, lightlike if equality holds and spacelike otherwise.

The 4-volume enters the Regge action with a cosmological constant and is the discrete analogon of the volume element $\mathrm{d}^4x\,\sqrt{-g}$,

$$V_{(4)}^2 = \frac{(j_0 + j_1)^2}{32}\left(8\underline{k}^2 + (j_0 - j_1)^2\right)\,. \tag{A4}$$

In this work, we focus exclusively on the causally regular sector, that is, timelike trapezoids, $\underline{k}^2 > 0$. The two types of deficit angles, those located at trapezoids and those located at squares, are respectively given by

$$\delta^{\mathrm{E}} = \frac{\pi}{2} - \cos^{-1}\left(\frac{(j_n - j_{n+1})^2}{16k^2 + (j_n - j_{n+1})^2}\right)\,, \tag{A5}$$

and

$$\delta^{\mathrm{L}} = \sinh^{-1}\left(\frac{j_1 - j_0}{\sqrt{16k^2 + (j_n - j_{n+1})^2}}\right)\,. \tag{A6}$$

δ^{E} is a Euclidean angle since the orthogonal subspace to a timelike trapezoid is $\mathbb{R}^2$. In contrast, δ^{L} is a Lorentzian angle as the orthogonal subspace to a spacelike square is $\mathbb{R}^{1,1}$.

Let us repeat here the Regge action of the causally regular sector,

$$S_{\mathrm{III}}^{(n)} = 6(j_n - j_{n+1})\sinh^{-1}\left(\frac{j_{n+1} - j_n}{\sqrt{16k_n^2 + (j_n - j_{n+1})^2}}\right) + 12k_n\left[\frac{\pi}{2} - \cos^{-1}\left(\frac{(j_n - j_{n+1})^2}{16k_n^2 + (j_n - j_{n+1})^2}\right)\right] - \Lambda V^{(4)}\,. \tag{A7}$$

The equations of motion are given by

$$\frac{\partial S_{\text{III}}}{\partial k_n} = 0\,, \quad \frac{\partial S_{\text{III}}}{\partial j_n} = 0\,, \tag{A8}$$

which can be simplified using the Schläfli identity and the additivity of the action. That is, the above equations can be reformulated as

$$\frac{\partial S^{(n)}_{\text{III}}}{\partial k_n} = 0\,, \quad \frac{\partial S^{(n-1)}_{\text{III}}}{\partial j_n} + \frac{\partial S^{(n)}_{\text{III}}}{\partial j_n} = 0\,, \tag{A9}$$

with the derivative terms explicitly given by

$$\begin{aligned} \frac{\partial S^{(n)}_{\text{III}}}{\partial k_n} &= 12\left[\frac{\pi}{2} - \cos^{-1}\left(\frac{(j_n - j_{n+1})^2}{16k^2 + (j_n - j_{n+1})^2}\right)\right] - \Lambda\frac{(j_n + j_{n+1})^2 k_n}{4V_{(4)}}\,, \\ \frac{\partial S^{(n)}_{\text{III}}}{\partial j_n} &= 6\sinh^{-1}\left(\frac{j_{n+1} - j_n}{\sqrt{16k_n^2 + (j_n - j_{n+1})^2}}\right) - \Lambda\frac{j_n(j_n - j_{n+1}) + 4k_n^2}{2\sqrt{(j_n - j_{n+1})^2 + 8k_n^2}}\,. \end{aligned} \tag{A10}$$

Appendix B: Cosmology in the continuum

The line element of a spatially flat, homogeneous and isotropic universe is given by

$$\mathrm{d}s^2 = N^2(t)\,\mathrm{d}t^2 - a^2(t)\,\mathrm{d}\mathbf{x}^2\,, \tag{B1}$$

with $a(t)$ the scale factor and $\mathrm{d}\mathbf{x}^2$ the 3-dimensional Euclidean line element. The lapse function, $N(t)$, corresponds to a choice of gauge emerging from the time reparametrization invariance of the system.

Dynamics of the cosmological system are governed by the Friedmann equations which, in the case where matter only consists of a cosmological constant, are simply given by

$$\left(\frac{\dot a}{a}\right)^2 = N^2\frac{\Lambda}{3}\,. \tag{B2}$$

This is straightforwardly solved by

$$a(t) = a_0 \exp\left(\pm\frac{\Lambda}{3}\int_{t_0}^{t} \mathrm{d}t'\, N(t')\right)\,, \tag{B3}$$

where we identify the gauge-invariant elapsed proper time as

$$\tau^{\text{cont}}(t_0, t) = \int_{t_0}^{t} \mathrm{d}t'\, N(t')\,. \tag{B4}$$

The proper time plays an important role in the comparison between discrete and continuum equations as the cumulative height of frusta converges to the proper time in an appropriate continuum limit.

In the discrete, as well as for effective spin-foams in ensuing studies, one is not given an initial scale factor and an initial time but rather an initial and final scale factor (corresponding to $\sqrt{j_0}$ and $\sqrt{j_{\mathcal{V}_T}}$, respectively). If, in the continuum, a_0, a_1 and Λ are given, the proper time is fixed to

$$\tau_0^{\text{cont}} = \log\left(\frac{a_1}{a_0}\right)\sqrt{\frac{\Lambda}{3}}\,. \tag{B5}$$

This quantity serves as a limiting value of $\tau = \sum_n H_n$ in the continuum limit described in Sec. V.

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